DjangoSentinel Developer Manual

Architecture, extension points, and contribution guide

DjangoSentinel v1.0.0 — Developer Manual

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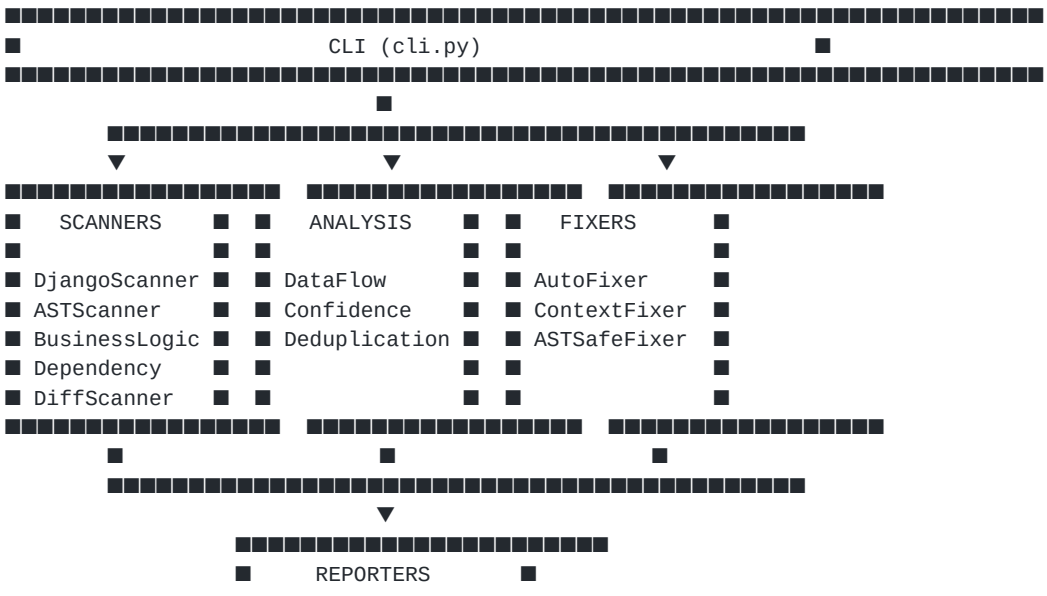
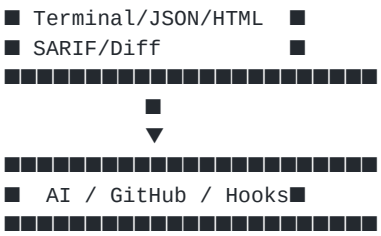
Architecture Overview

DjangoSentinel is a multi-engine security scanner built on a pipeline architecture:





Engine Layers:

1. **Rule Engine** — regex + AST patterns (always runs)
2. **AST Engine** — structural code analysis (always runs)
3. **Taint Engine** — data-flow source→sink tracking (always runs)
4. **AI Engine** — Claude-powered fixes/explanations (`--ai`)
5. **Diff Engine** — git-diff scoped scanning (`--diff`)
6. **Cache Engine** — file-hash incremental scanning (automatic)

Project Structure

```
django-sentinel/
├── djangosentinel/
│   ├── __init__.py          # Package exports & version
│   ├── __main__.py          # python -m entry point
│   ├── cli.py               # CLI parser & command handlers
│   ├── models.py            # Core data models
│   ├── cache.py             # Scan cache & incremental scanner
│   ├── ai_engine.py         # Claude AI integration
│   ├── github_integration.py # GitHub PR API
│   ├── scanners/
│   │   ├── __init__.py
│   │   ├── django_scanner.py         # Django-specific patterns
│   │   ├── ast_scanner.py            # AST structural analysis
│   │   ├── business_logic_scanner.py # Logic flaws & IDOR
│   │   ├── dependency_scanner.py     # CVE database
│   │   └── diff_scanner.py           # Diff parsing & scoped scan
│   ├── analysis/
│   │   ├── __init__.py
│   │   ├── data_flow.py      # Taint tracking
│   │   └── confidence.py     # Scoring & deduplication
│   ├── fixers/
│   │   ├── __init__.py
│   │   ├── auto_fixer.py     # Rule-based fixer
│   │   ├── context_fixer.py  # Context-aware fixer
│   │   └── ast_safe_fixer.py # Indentation-preserving fixer
│   ├── reporters/
│   │   ├── __init__.py
│   │   └── reporters.py      # All output formats
│   └── hooks/
│       ├── __init__.py
│       └── git_hooks.py      # Git hook installer
├── tests/
│   └── test_all.py           # 104-test suite
├── docs/                     # Additional documentation
├── setup.py                  # Package metadata
└── pyproject.toml            # Build & tool config
```

Core Data Models

Located in `djangosentinel/models.py`.

`Severity`

Enum: `CRITICAL`, `HIGH`, `MEDIUM`, `LOW`, `INFO`

`VulnCategory`

Enum: `INJECTION`, `XSS`, `CSRF`, `AUTH`, `SENSITIVE_DATA`, `MISCONFIG`, `INSECURE_DEPS`, `BUSINESS_LOGIC`, `IDOR`, `SSRF`, `XXE`, `OPEN_REDIRECT`, `RACE_CONDITION`, `MASS_ASSIGNMENT`, `HARDCODED_SECRET`

`CodeLocation`

```
@dataclass
class CodeLocation:
    file_path: str
    line_number: int
    column: Optional[int] = None
    end_line: Optional[int] = None
    snippet: Optional[str] = None
```

`Fix`

```
@dataclass
class Fix:
    description: str
    original_code: str
    fixed_code: str
    applied: bool = False
    manual_review_required: bool = False
    explanation: str = ""
```

`Vulnerability`

```
@dataclass
class Vulnerability:
    vuln_id: str
    title: str
    category: VulnCategory
    severity: Severity
    location: CodeLocation
    description: str
    owasp_ref: str = ""
    cwe_id: str = ""
    cve_id: str = ""
    fix: Optional[Fix] = None
    false_positive: bool = False
    references: List[str] = field(default_factory=list)
    metadata: Dict[str, Any] = field(default_factory=dict)
```

Key properties: `confidence`, `confidence_label`, `is_taint_detected`, `is_ai_detected`, `is_ast_detected`

`ScanResult`

Aggregates findings with counts: `critical_count`, `high_count`, `medium_count`, `low_count`, `fixable_count`, `fixed_count`, and `summary()`.

Scanner Development

All scanners follow the same interface:

```
class MyScanner:
    def __init__(self, project_path: str):
        self.project_path = Path(project_path)
        self.vulnerabilities = []

    def scan(self) -> List[Vulnerability]:
        # Walk files, detect patterns, return findings
        return self.vulnerabilities
```

Adding a Scanner

1. Create `djangosentinel/scanners/my_scanner.py`
2. Implement `__init__(self, project_path)` and `scan() -> List[Vulnerability]`
3. **Always set** `metadata["source"]` — used by confidence scoring:

   ```
   v.metadata["source"] = "my_scanner"
   ```

4. Export it in `scanners/__init__.py`
5. Register it in `cli.py` `run_full_scan()`:

   ```
   scanners_to_run = [
       ("■ Django Scanner", lambda: DjangoScanner(project_path).scan(), True),
       ("■ My Scanner",     lambda: MyScanner(project_path).scan(),     True),
   ]
   ```

Base confidence values (`analysis/confidence.py`)

```
BASE_CONFIDENCE = {
    "ast_scanner":        0.82,
    "taint_tracker":      0.90,
    "django_scanner":     0.75,
    "business_logic":     0.70,
    "dependency_scanner": 0.95,
    "ai_pattern_scanner": 0.65,
    "diff_scanner":       0.75,
    "unknown":            0.60,
}
```

AST Scanner Example

Use `ast.parse()` and `ast.NodeVisitor`. See `scanners/ast_scanner.py` for the visitor pattern with parent tracking.

Diff Scanner

`DiffScanner` wraps other scanners and filters findings to changed lines only. See `DiffParser` for unified diff parsing (`all_added_lines`, `hunks`, `is_new_file`).

Fixer Development

`AutoFixer`

Rule-based fixer that delegates to `ASTSafeFixer`. Creates backups in `.sentinel_backups/`.

`ContextAwareFixer`

Combines rule-based fixes with AI. Categories:

1. Rule-fixable (`fix` present, `manual_review_required=False`, `fixed_code` not starting with `#`)
2. AI-needed (`manual_review_required=True` or placeholder hint)

3. Manual (no fix)

`ASTSafeFixer`

The core fix application engine. Four-stage approach:

1. **Exact inline replacement** — single line, simple swap
2. **Line-number targeted block replacement** — multi-line with indentation preserved
3. **Pattern-based replacement** — using `location.snippet`
4. **Whole-source replacement** — last resort

Critical invariants:

- Parse source with `ast.parse()` before touching it
- Validate `ast.parse(new_source)` after the fix — abort if syntax breaks
- Detect and preserve target line indentation via `detect_indent()` / `apply_indent()`
- Never apply fixes with `manual_review_required=True`

Adding a rule-based fix: add `fix_original` / `fix_replacement` to a check dict in `DjangoScanner.SETTINGS_CHECKS`.

Reporter Development

All reporters implement `report(self, result: ScanResult) -> str`.

```
class MyReporter:
    def report(self, result: ScanResult) -> str:
        return "..."
```

Register in `cli.py` `cmd_scan()` and export in `reporters/__init__.py`.

`DiffReporter`

Specialised for `--diff` mode. Takes `diff_summary` dict and implements both `report()` (terminal) and `report_json()`.

`SARIFReporter`

Produces SARIF 2.1.0 for GitHub Code Scanning. Maps severity to SARIF levels (`error`/`warning`/`note`).

Analysis Engine

Data-Flow Taint Tracking (`analysis/data_flow.py`)

```
tracker = DataFlowTracker(project_path)
flows = tracker.analyze_project()   # all files
flows = tracker.analyze_file("views.py", source)  # single file
```

Concepts:

- `TAINT_SOURCES` — where user data enters (`request.GET`, `request.POST`, etc.)
- `SINKS` — dangerous operations (dict: pattern → `{vuln, cwe, severity}`)
- `SANITIZERS` — functions that clean tainted data (`int()`, `escape()`, `get_object_or_404`)
- `_TaintVisitor` — AST visitor tracking taint propagation

- `DataFlow` dataclass — confirmed source→sink flow

Adding a sink: add to the `SINKS` dict:

```
"my_dangerous_call": {"vuln": "My Vuln", "cwe": "CWE-XXX", "severity": "HIGH"},
```

Confidence Scoring (`analysis/confidence.py`)

- `compute_confidence(vuln, source_line)` — base score + severity boost + signal checks
- `deduplicate(vulns)` — merges findings by `finding_hash` (file + line bucket + category + CWE)
- `apply_confidence_scores(vulns, source_map)` — bulk scoring
- `filter_by_confidence(vulns, min_confidence)` — threshold filter
- `score_label` / `score_emoji` — display helpers

Deduplication: keeps highest-confidence finding; merges references, metadata, and adopts a fix from duplicates if the winner lacks one.

AI Engine

Located in `ai_engine.py`. Pure HTTP (no SDK dependency).

```
_call_claude(system, user, max_tokens, api_key, timeout) -> Optional[str]
```

Classes:

- `AIFixEngine` — generates fixes for `manual_review_required` vulns
- `AIExplainer` — developer-friendly explanations
- `AIPatternScanner` — reasons about novel patterns (`--ai-deep`)
- `PRReviewSummarizer` — GitHub PR markdown summaries

All AI classes expose `.available` (True if `ANTHROPIC_API_KEY` set) and are no-ops when unavailable.

Adding an AI capability: add a class with a `SYSTEM_PROMPT`, call `_call_claude()`, parse the response, and return a `Fix` / `str` / `List[Vulnerability]`.

GitHub Integration

Located in `github_integration.py`.

```
gh = GitHubPRIntegration(token=..., repo=..., pr_number=..., commit_sha=...)
gh.available                      # bool
gh.post_summary_comment(result, summarizer)
gh.post_inline_review(result)     # -> count
gh.create_check_run(result)       # -> bool
gh.get_pr_changed_files()         # -> List[str]
```

Environment variables auto-detected: `GITHUB_TOKEN`, `GITHUB_REPOSITORY`, `GITHUB_SHA`, `GITHUB_REF`. PR number parsed from `refs/pull/N/merge`.

Cache System

Located in `cache.py`.

- `file_hash(path)` — SHA-256 (first 16 chars)

- `ScanCache` — persistent JSON cache at `.sentinel_cache/scan_cache.json`
- `IncrementalScanner` — wraps any scanner with cache hits/misses

Cache rules:

- Only caches findings with `confidence >= 0.5`
- Never caches AI findings (title starts with `[AI]`)
- Invalidates entire cache on `SENTINEL_VERSION` change
- TTL: 7 days
- Conservative: cache-miss means re-scan

Adding cache invalidation: bump `SENTINEL_VERSION` in `cache.py`.

Testing

Run the full suite:

```
python -m pytest tests/test_all.py -v
# or
python tests/test_all.py
```

104 tests across 12 classes:

Class	Tests	Coverage
`TestDjangoScanner`	20	settings, views, templates, secrets
`TestASTScanner`	10	structural detections
`TestBusinessLogicScanner`	12	logic flaws + AST IDOR
`TestDependencyScanner`	6	CVE database, imports
`TestDiffParser`	8	unified diff parsing
`TestDiffScanner`	9	diff-scoped detection
`TestDataFlowTracker`	8	taint analysis
`TestConfidenceScoring`	8	scoring + dedup
`TestASTSafeFixer`	8	fix application
`TestScanCache`	6	file-hash cache
`TestAutoFixer`	3	simple fixes
`TestReporters`	5	output formats

Writing a Test

Use `TempDirMixin` for temp project directories:

```
class TestMyScanner(TempDirMixin, unittest.TestCase):
    def scan(self):
        return MyScanner(self.test_dir).scan()

    def test_detects_something(self):
        self.write_file("views.py", "dangerous_code()\n")
```

```
        self.assertTrue(any("Vuln" in v.title for v in self.scan()))
```

AI Model Provider Abstraction

DjangoSentinel uses a model provider abstraction layer defined in `djangosentinel.ai_model_providers`. This layer provides a uniform interface over multiple LLM providers while maintaining full backward compatibility.

Provider Classes:

- `AnthropicProvider` — wraps Anthropic Claude API
- `OpenAIProvider` — wraps OpenAI GPT API
- `GeminiProvider` — wraps Google Gemini API

Base Class: `AIModelProvider`

Each provider implements:

- `_get_api_key()` — retrieves API key from environment
- `_call_api(system, user, max_tokens, timeout)` — makes the API call
- `generate(system, user, max_tokens, timeout)` — public entry point

Global Provider Management:

- `get_ai_provider()` — returns the current default provider
- `set_ai_provider(provider)` — sets a new default provider
- `is_ai_available()` — checks if the current provider has an API key

Default Behavior:

The default provider is `AnthropicProvider`, which preserves full backward compatibility.

Existing code using `_call_claude`, `AIFixEngine`, `AIExplainer`, etc. continues to work

without any modifications.

Adding a New Provider:

To add support for a new LLM provider:

1. Create a class that inherits from `AIModelProvider`
2. Implement `_get_api_key()` to read the appropriate environment variable
3. Implement `_call_api()` with the provider's HTTP API details
4. Optionally register the provider or make it selectable via `set_ai_provider()`

Example: Custom Provider

```
from djangosentinel.ai_model_providers import AIModelProvider

class CustomProvider(AIModelProvider):
    def _get_api_key(self):
        return os.environ.get("CUSTOM_API_KEY", "")

    def _call_api(self, system, user, max_tokens, timeout):
        # Custom API call implementation
        pass
```

CLI Integration:

All CLI commands that use AI (`--ai`, `--ai-deep`, `--ai-explain`) respect the

currently set provider. No code changes are needed to support multiple providers

— just set the appropriate environment variable.

Docker Containerized Deployment and Configuration

DjangoSentinel ships with a multi-stage `Dockerfile` and a `docker-compose.yml`. They give the project two switchable environments built from the same source.

Dockerfile

Target	Purpose	Contents
`production` (default)	Run scans	`python:3.12-slim`, `git` (for `--diff`), DjangoSentinel installed from the local source, non-root user `sentinel`, `ENTRYPOINT ["sentinel"]`, `CMD ["scan", "."]`
`development`	Work on DjangoSentinel	Same base, `pip install -e ".[dev]"`, tests copied in, default command `python -m pytest -q`

The image is built from the local source, not from PyPI, so it always matches the checked-out code.

docker-compose.yml

- `sentinel` (production target): mounts `${SCAN_PATH:-.}` at `/src`, passes the AI and GitHub environment variables through, `restart: "no"`.
- `dev` (development target, profile `dev`): mounts the source at `/opt/sentinel` for live editing and the project to scan at `/src`.

```
docker compose run --rm sentinel scan .                    # production environment
docker compose --profile dev run --rm dev                  # development: run tests
docker compose --profile dev run --rm dev bash             # development: shell
```

Environment file

Copy `.env.example` to `.env`. Provide one AI key (`ANTHROPIC_API_KEY`, `OPENAI_API_KEY` or `GOOGLE_API_KEY`) and, for pull-request features, the GitHub variables. Keys are read at run time and never stored in the image.

CI usage

For GitHub Actions or other CI systems, use the environment variables directly without Docker Compose (see `docs/CI_INTEGRATION.md`).

Contributing

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/my-scanner`)

3. Add your code following existing patterns
4. Add tests (maintain 104+ passing)
5. Run `python -m pytest tests/test_all.py -v`
6. Ensure no regressions
7. Submit a pull request

Code Style

- No comments unless they explain non-obvious logic
- Type hints on public functions
- Dataclasses for data structures
- Emoji icons in terminal output for user-friendliness
- Errors caught gracefully (`except Exception: continue`) in scanners to never crash a scan

Architecture Principles

1. **Scanners are dumb** — detect and return, no side effects
2. **Analysis is pure** — scoring/dedup take lists, return lists
3. **Fixers are safe** — validate before and after; never corrupt code
4. **Reporters are pure** — take `ScanResult`, return string
5. **AI is optional** — every AI feature has a non-AI fallback

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Version History

- **1.0.0** — Initial stable release: full scanner suite, taint analysis, auto-fixers, 5 reporters, AI engine, GitHub integration, git hooks, incremental cache.

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